

Machine Learning Enabled System Architecture for Automatic Grading and Sorting of Walnut Fruits: A Review

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Abstract— India is the eighth largest walnut producer worldwide and exports walnuts to 70 countries worth 10 million USD (778 million INR) annually. The grading of walnut ripeness is the main concern of worry for customers and the fruit business. The efficacy and fast classification of a walnut's ripeness stage are crucial in judging the quality. Walnut maturity classification mainly influences walnut fruit quality and later marketing. Our approach to the automatic sorting and grading of walnut ripeness focuses on the identification of marks that appear on the outer surface of walnut fruits. The image dataset for the identification of marks or patches is divided into four different classes unripe, mid-ripe, ripe, and overripe acquired through high-quality cameras used in the proposed architecture, performing feature engineering and classification using ANN, SVM, and CNN techniques. The Raspberry Pi function counts fruits in a fruit box, using a conveyor belt to detect marks and patches for grading and sorting of walnut fruit based on texture, size, and color features. A motor powers the belt, moving walnuts from the fruit placer to the mechanical part. This architecture automates the fruit grading process, significantly reducing labor costs and time compared to manual sorting. This improves productivity and allows for faster processing of large quantities of fruits and is capable of quality control, increased efficiency, reduced errors, and improved market access.

Keywords— walnut ripeness classification, image processing, machine learning, fruit grading, sorting, artificial neural network, svm, cnn

I. INTRODUCTION

Walnut is widely known as “akhroot or akhrot” in India. Walnut is one of the temperate nut fruits that is associated with the variety of Juglandaceae [1]. A Walnut's outer shell is hard and there are seeds within. It's a nut type of fruit. walnut is grown at a sea level of 900-3000 meters [2]. In India, walnuts are grown in the northwest Himalayan region to the north-eastern region, due to Crop-specific climate conditions, the Himalayan range is suitable for walnut crops, which ripen between September and November in the northern hemisphere. There are four major walnut fruit

cultivator states in India, Jammu and Kashmir, Himachal Pradesh, Maharashtra, and Uttaranchal are the main walnut producers in India [3]. Even so, walnut farming in Arunachal Pradesh has rapidly increased in recent decades and other states are also associated with commercial walnut farming. Jammu and Kashmir is the main walnut-producing region in our country, especially Kashmir Valley. The overall area of walnut land in India is 80,800, hectares which produce 102,200Mt, production of walnuts annually. India exports walnuts to almost 70 countries and produces around 10 million USD per year. Universally, India is ranked eighth in walnut production around the globe [4]. The production of walnut crops has increased in recent years from 185 to 299 thousand metric tonnes and the area also increased from 75 to 108 thousand hectares between 2000-2018 according to Horticulture at a Glance, 2018, Ministry of Agriculture report, Government of India [5].

The main problem that farmers face while walnut farming is a physical classification of walnut ripeness with one's hands. Manually classification of walnut ripeness is tiresome work, a very time-consuming process, and increases labor costs, so an automatic sorting and grading platform is required based on machine learning, which saves manpower, time, and also cost. Predominantly, used features for walnut maturity classification are the color, texture, size, and shape of walnut shell for identification of ripeness [6]. Among all these attributes husk color of the walnut is the major parameter that decides the ripeness of the walnut. The outer layer of an immature walnut is firm, and green, and sticks tightly to the nut. The walnut fruit ripeness stages based on husk color are classified into four categories for classifying the ripeness using machine learning techniques as shown in Fig. 1.

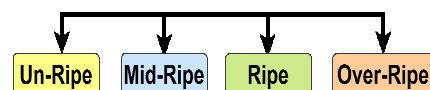


Fig. 1. Walnut maturity level classification.

In this research paper, we introduce an automatic architecture for walnut fruit grading and sorting based on classification algorithms using image processing to classify the ripeness of walnut fruit premised on the melding of color and texture features of walnut husk. The classification algorithms used in this proposed architecture are ANN, SVM, and CNN algorithms. This proposed architecture helps in the automatic classification and grading of walnut fruits and reduces manpower and labor costs.

II. GENERAL FRUIT RIPENESS DETECTION AND CLASSIFICATION SYSTEM

Through digital image processing, the ripeness of the fruits may be determined based on their texture, shape, size, color, and morphological properties. The image processing technique involves five key steps, and the data flow diagram is given below in Fig. 2.

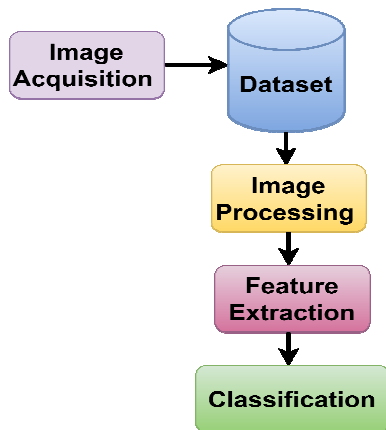


Fig. 2. Fruit ripeness classification system.

A. Image Acquisition

Image acquisition is the initial step of a walnut fruit ripeness classification system. One can obtain high-quality crop images through the use of digital cameras, scanners, or drones. Acquired images can be annotated to display data features so that the model can be on its own. This can be done by labeling or classifying the image using language, annotation tools, or both.

B. Image Processing

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C. Feature Extraction

The concept of image preprocessing describes operations performed on images at the lowest probable degree of abstraction to improve image quality and get it ready for additional processing. A thorough preprocessing step can greatly enhance the results of later image analysis assignments. It involves identifying and describing local features or patterns in an image, which are essential for tasks like object detection, image matching in fruit grading and sorting applications, and image recognition. Fruit image

feature extraction can be accomplished through the use of machine intelligence, Blend vision, color, shape, and grey-level Co-occurrence matrix (GLCM) texture features

D. Classification

One critical technological advancement that improves the productivity and precision of the agriculture sector is image classification in fruit grading and sorting. The agriculture sector is highly dependent on the caliber and reliability of its output. Fruit grading and sorting by hand using traditional methods are costly, time-consuming, and vulnerable to human mistakes. Image categorization technology ensures high accuracy, and consistent quality standards, and automates these operations, making it a potential solution. Ultimately, any machine learning technique can be used to classify fruit maturity level whether a fruit is ripe, unripe, or overripe.

III. RELATED WORK

Numerous approaches and strategies have been used in reviewed literature to grade the maturity of the fruit, using parameters such as size, color, texture, and morphology of fruits.

A framework based on computational intelligence is used for the identification of ripe and unripe Grape clusters in two steps, segmentation separates the grape bunch and the backdrop Color of the image and in the next Stage, the ripe grape and unripe grape bunch will be separated. In this framework, ripeness is classified using computational intelligence and image processing techniques. A dataset of 31 images of grapes with 2248×3264 resolution was used and achieved 96.88% accuracy using the SVM algorithm [7]. For pineapple maturation, a fuzzy logic approach has been suggested. It uses characteristics retrieved from the color channel and CIE Lab. 93.11% efficacy was attained by utilizing the model [8]. Banana ripeness is classified using two algorithms, namely Area element technique and degree of color, and classification is done with GLCM achieved with 99.1% accuracy using ANN. The main drawback of area feature algorithms is that they cannot accurately classify ripe and over-ripe bananas [9]. A system designed to classify mango fruit is introduced which classifies the ripeness of mango into four stages based on ripeness level, In this research paper for maturity classification, a video of mango fruit is captured with a camera from a mango garden. 27 types of features are extracted from captured video signals. Later on, support vector machine and recursion feature techniques are applied for grouping purposes. 16,400 total images are used and obtained with 96% accuracy [10]. A sensor-based approach is used in the classification of oil Palm which detects ripeness based on spectral bands and achieves 85% accuracy [11]. Another model is introduced, which estimates ripeness in strawberries drawing from images with multiple wavelengths. In this model Support vector machine (SVM) and PCA with BPNN, achieved an accuracy of 100% for strawberry fruit, ripeness classification Factors such as dimensions, hue, and roughness are utilized for ripeness classification. Additionally, the analysis uses the RGB color feature, which is an authentic feature over color, size, and texture. The result between image analysis and correlation was 99.66% accuracy [12].

Numerous research works have been reported over a few years related to the fruit market in different aspects such as disease detection fruit ripeness estimation, classification based on vision systems, ML, and AI techniques. Despite of fruit industry, the application of vision systems is significantly used by the agriculture sector. Walnut fruit achieves 85% accuracy. Another model is introduced, grading and sorting using machine learning techniques there are a few publications available such as a yoloV4 and Mobile NetV3 model in which feature is extracted based on ripeness a total of 3600 jpg format walnut images of 3240×4032 pixels as samples are used. 2880 images are used in the training set, 320 in the validation set, and 400 for the testing set. The features extracted such as size, color, and texture based on the ripeness category of walnut fruit using RGB. The YOLOv4 model prediction for the training set is 84.5%, for the validation set YOLOv4 model gives 90.72% accuracy. For the testing set the accuracy of the YOLOv4 model is 76.92% [13]. To understand the existing work on fruit grading and sorting a comparative analysis of the present techniques with their accuracy is discussed in Table 1. and fig.3.

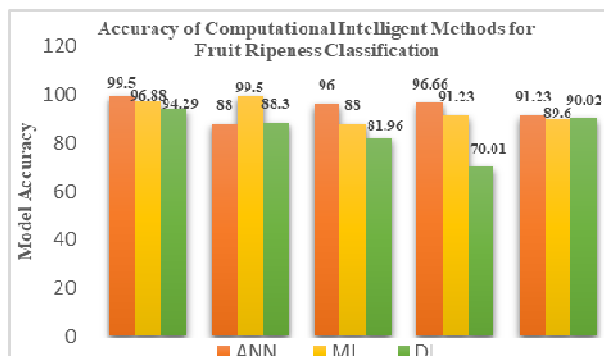


Fig. 3. Existing computational techniques for fruit ripeness classification with accuracy.

IV. PROPOSED SYSTEM ARCHITECTURE

ANN, SVN, and CNN are used in this proposed system's methodology for sorting and grading walnut fruits using their color, shape, and texture characteristics. This proposed architecture performs the vision perceptivity like a human operator in making decisions at a very fast rate. Our approach to the sort and grade walnut maturity is based on the identification of marks that appear on the outer surface of walnuts. In this proposed automatic architecture system as shown in Fig. 5, the dataset of walnut images is divided into four different classes—unripe, mid-ripe, ripe, and overripe—acquired through a high-quality camera sensor, performs feature engineering operations for the training and testing of walnut fruit as shown in Fig. 5 and Fig. 6. The walnut fruit image block's features include weights, sizes, shapes, and colors that indicate the fruit's quality. This block feeds the algorithm's processing block with feature data as well as training and test photos of walnut fruits. The total number of fruits in the fruit box is counted using the Raspberry Pi function. Walnut fruits are captured by Raspberry Pi cameras, and then computational methods for image processing evaluate the images to determine how good they are for sorting and grading. Initially, fruit is placed in a fruit placer, and a rolling machine which is connected to an electric motor performs the function of moving fruits towards the mechanical part where grading and sorting can be done. The Raspberry Pi device is connected to a

mechanical part as shown in Fig. 5 below which counts a total number of fruits and displays the result. This analysis considers features based on texture, size, color, and shape. Fruit grading can be identified and categorized by grading and sorting blocks according to maturity levels using predictive algorithms that have been trained on walnut fruit image datasets. This allows the fruit sorting system to distinguish between good and bad fruit classes.

A computational algorithm is used for matching features function to compare the camera-captured images to determine whether walnut fruit is ripe, unripe, mid-ripe, or overripe, the algorithm uses this block to analyze size, husk texture, color, and weight. It then uses grading and sorting blocks to classify the walnut ripeness as shown in Fig. 6.

A. Dataset

There are varying numbers of walnut fruit images in the dataset. Images are split by several factors based on their ripeness levels such as unripe, ripe, mid-ripe, and overripe based on walnut size, weight, shape, and color features as shown in Fig. 4.

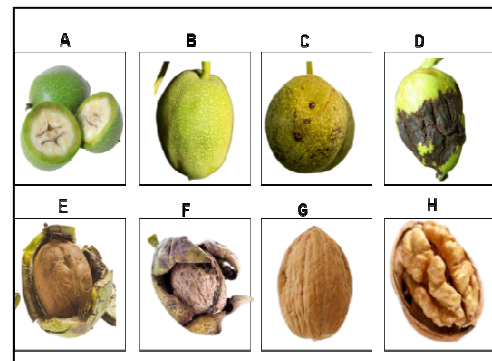


Fig. 4. Samples of different walnut fruits. A is for green un-ripe walnut B, C, and D are for the yellowish-green mid-ripe walnut, E, F (open husk with shell) are for the ripe walnut and G, H (outer shell texture) are for the overripe walnut.

B. Learning Algorithms

- Artificial Intelligence:** Three distinct layers compose an ANN an input layer, a hidden layer, and an output layer. The neurons are linked to parameters of bias function and weights. The hidden layers and prejudice purposes depend on the type of ANN applied in the system. Weights are used to store information about the input data, and each neuron performs the summation function. On error occurrence, the output network feeds the information back to the input via a backpropagation connection.
- Support Vector Machine:** SVM learners find the most suitable line or hyperplane in an N-dimensional space to maximize the distance between each class and classify walnut fruit grading and sorting. The architecture of SVM is described in Figs. 7 and 4. Often known as eigenvalues manifolds, surfaces, subspaces, and polynomials are flat surfaces., etc. Images of walnuts are classified according to the type of SVM that is being utilized. The vector machine's plane surfaces represent the boundary of positive and negative support vectors, determined by features in the dataset's images. working of ANN and SVM are shown in Fig. 7.

TABLE I. COMPARATIVE ANALYSIS OF FRUIT RIPENESS BASED ON IMAGE PROCESSING AND COMPUTATIONAL TECHNIQUES APPROACHES.

Author and Year	Fruits	Image Processing and Computational Intelligence Methods for Fruit Ripeness Classification			
		Artificial Neural Network	Deep Learning	Machine Learning	Results
Zuñiga et al. 2014 [14]	Mango	✗	✗	SFFS and SVM	96% Accuracy
Goel et al. 2015 [15]	Grapes		Fuzzy logic		94.29% Accuracy
Alireza Sanaeifar et al. 2016 [16]	Tomatoes ripeness estimation	Artificial Neural Networks	✗	Support Vector Regression	99.50% Accuracy
Prabira Kumar Sethy et al. 2017 [17]	Banana	Artificial Neural Networks	✗	KNN, SVM, and FUZZY LOGIC	88% Accuracy
Raja Hamza et al. 2018 [18]	—	Artificial Neural Networks	✗	✗	96.66% Accuracy
Hzzhu et al.2020 [19]	Apple	Artificial Neural Networks	✗	Stochastic Gradient Boosting	91.23% Accuracy
Nor-Aziana Aliteh et al. 2020 [20]	Oil Palm	Artificial Neural Networks	✗	✗	90.4% Accuracy
Cui, Kaixuan et al. 2022 [21]	✗	CNN	✗	✗	88.30% Accuracy
Worasawate, Denchai et al.2022 [22]	Walnut	✗	K-Nearest Neighbour	✗	89.6% Accuracy
Nagnath Aherwadi et al. 2022 [23]	Mango	CNN	✗	✗	81.96% Accuracy
Hanh et al. 2023 [24]	Apple	CNN	✗	✗	99.5% Accuracy
Tomas et al. 2024 [25]	Tomato	✗	CNN	✗	70.1% Accuracy
Shiddiq et al.2024 [26]	Oil Palm	ANN	✗	✗	90% Accuracy
Olisah et al. 2024 [27]	Blackberry	Ensemble CNN	✗	✗	90.2 % Accuracy

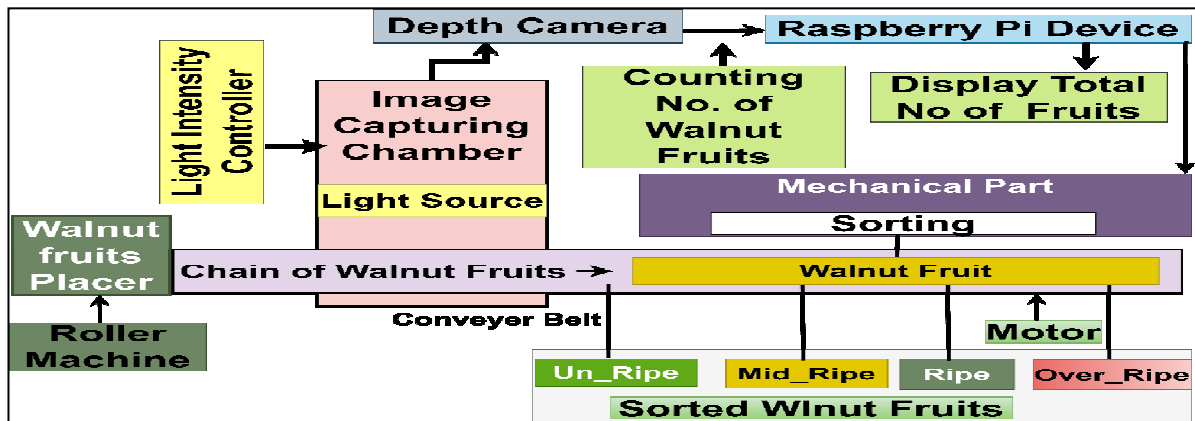


Fig. 5. Proposed system architecture for grading and sorting of walnut fruits

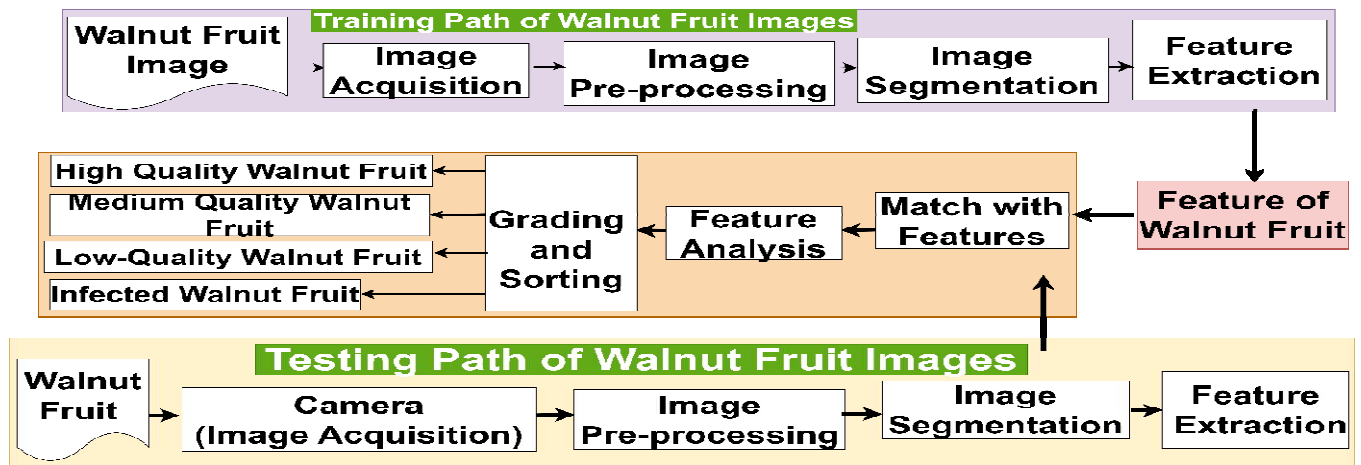


Fig. 6. The working path of the proposed system architecture for walnut fruit grading and sorting.

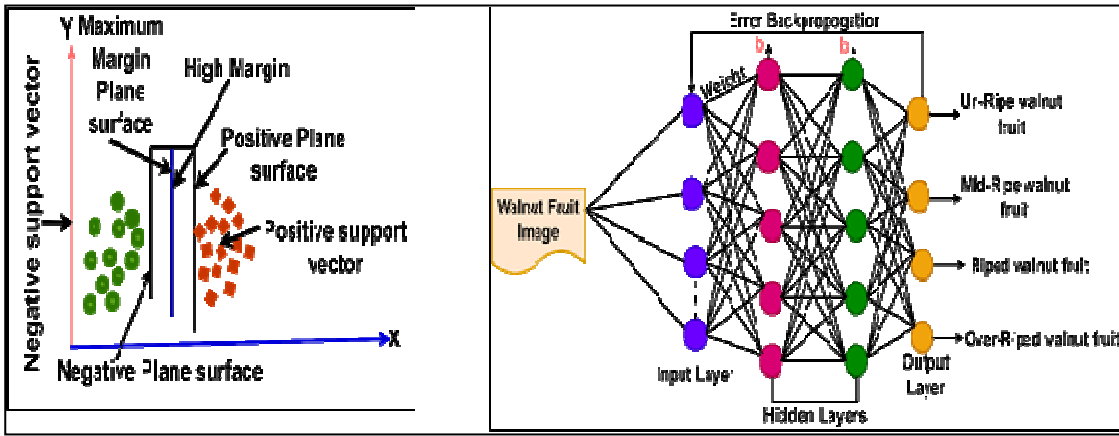


Fig. 7. Architecture and representation of the artificial neural network and support vector.

c) *Convolutional Neural Network (CNN)*: CNN is a framework that is applied in this study to perform the functionality of auto sorting and grading. As seen in Fig. 5, the design of the framework encompasses different phases in the algorithm's process of training. Four functional components are included throughout each phase: max pooling function (subsampling layer), linear rectified units (a kind of ReLU), convolution layer, filters bank, or kernels. These steps represent the feature-established array of feature maps. Fig. 8 depicts the fundamental architecture of CNN. This creates the network's ultimate output layer and connects all operating stages to form fully linked layers. The backpropagation network is coupled to an input route or layer, and the error function appears when an error arises in the output layer.

	correctly identify the data. It has been confirmed that Random Forest, SVM Classifier, Linear SVC, and Logistic Regression have the maximum accuracy. SVM and Random Forest, with a classification accuracy of 99.96%.
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V. LIMITATIONS OF EXISTING FRUIT GRADING AND SORTING SYSTEM

Farming has been transformed by automated fruit grading systems that use machine learning and the Internet of Things to maximize post-harvest activities, reduce labor costs, and increase crop output. Notwithstanding obstacles like initial outlay, infrastructure, and data security, the suggested system that makes use of AI and IoT can provide a more sustainable way of farming.

VI. CONCLUSION

Using features of the walnut fruit, machine learning algorithms including CNN, SVM, and ANN will be employed to identify and sort the walnut fruits. The basic principle of feature extraction consists of image processing techniques that enhance the quality of the imagery dataset. The machine learning algorithm's feature detection and classification accuracy is determined by the overall number of iterations used to train the network. The biased and weighted function of an ANN network must be updated as the hidden layer is updated. Due to the error function that occurs in ANN's output, there are growing amounts of repetitions. As a result, the backpropagation technique is used to reduce error and the output neuron generates output. When the values of the support vectors change in the SVM method, the eigenvectors' values also change. The CNN functions similarly to the ANN functions. Compared to ANN and SVM, the convolution function gives an effective output and speeds up processing. The CNN functions similarly to the ANN functions. Compared to ANN and SVM, the convolution function gives an effective output and speeds up processing.

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TABLE II. COMPARATIVE ANALYSIS BASED ON RASPBERRY PI FOR FRUIT GRADING AND SORTING.

Published Year	Published Year
Jayalakshmi et al. 2022 [28]	Using RASPBERRYPI/PYTHON, the investigation applies image processing to distinguish either normal or ripe fruits and vegetables. With its ability to precisely evaluate product quality and precision, this technology may eventually replace manual food inspection with an impartial, non-destructive grading system. This straightforward method ensures proper consumption and reduces food waste by assisting customers in assessing the quality of veggies and fruits before consuming them. By supplying fresh fruits and vegetables, the system improves individual health; it can be further improved by using the payload technique, which uses non-contact methods to correctly analyze the internal structure of fruits.
Kuo et al. 2022 [29]	Apples, lemons, and oranges fruits are detected using an SSD neural network model, and the Raspberry Pi is utilized for edge computing tasks such as AI identification, successfully identifying healthy and unhealthy fruits, and MQTT-based interaction and movement with an AGV automobile. Conveyor belts are utilized by the framework to move and classify quality fruit, eliminating ruined samples so that the fruit grading model loses just 1% of the all. The probability of correctly identifying a fruit during the phase of identification will be greater than 80%.
Sahu et al. 2020 [30]	Over a week, data was collected using the Raspberry Pi in conjunction with commercial outside sensors. After using the PCA technique to visualize the data, it was found that the data was not linearly separable. Consequently, machine learning models and both linear and non-linear classification techniques were used to

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